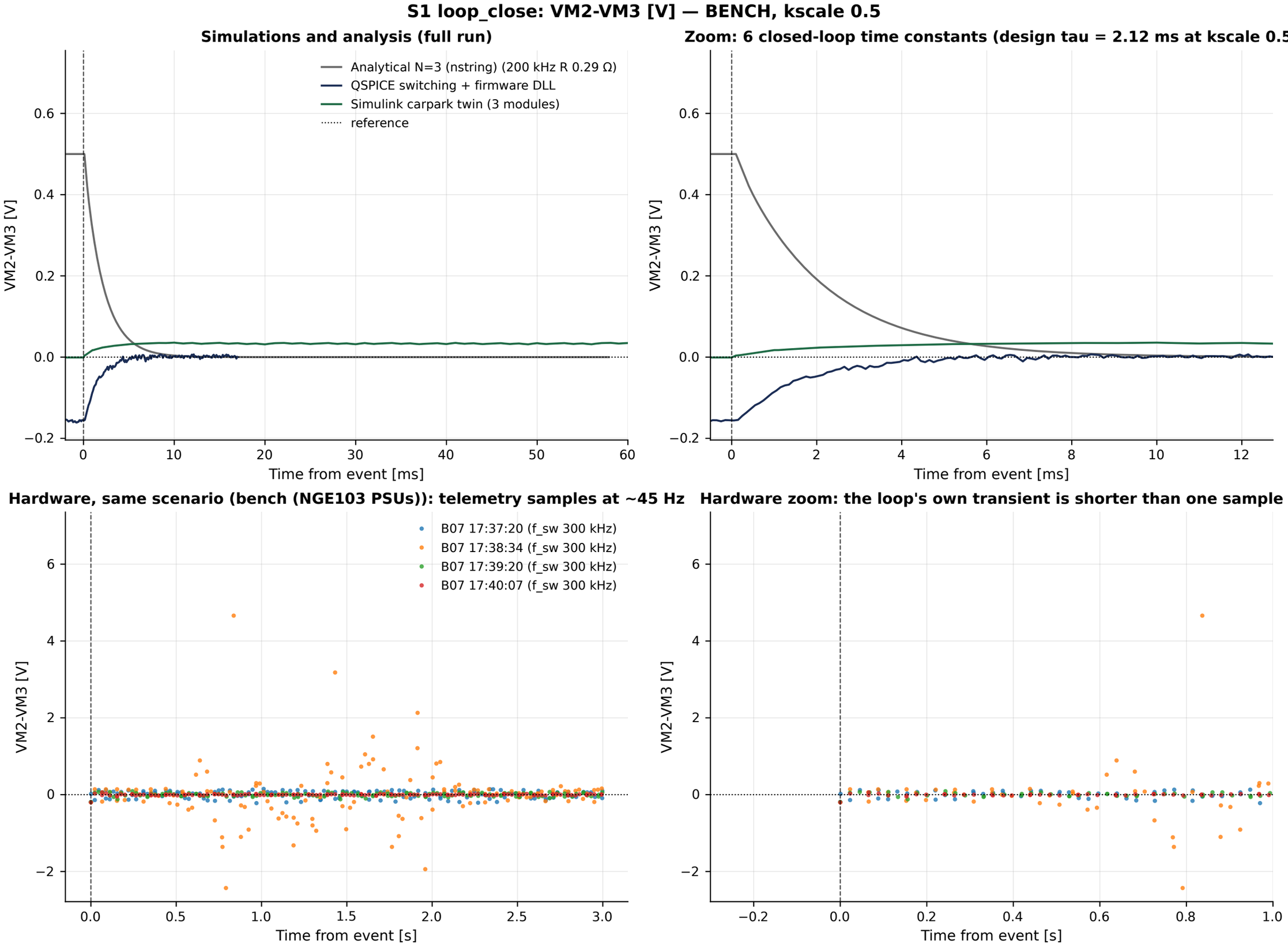
S1 loop_close: VM2-VM3 [V] — BENCH, kscale 0.5
Simulations and analysis (full run)
Analytical N=3 (nstring) (200 kHz R 0.29 Ω)
QSPICE switching + firmware DLL
Simulink carpark twin (3 modules)
reference
VM2-VM3 [V]
Time from event [ms]
Zoom: 6 closed-loop time constants (design tau = 2.12 ms at kscale 0.5
VM2-VM3 [V]
Time from event [ms]
Hardware, same scenario (bench (NGE103 PSUs)): telemetry samples at ~45 Hz
B07 17:37:20 (f_sw 300 kHz)
B07 17:38:34 (f_sw 300 kHz)
B07 17:39:20 (f_sw 300 kHz)
B07 17:40:07 (f_sw 300 kHz)
VM2-VM3 [V]
Time from event [s]
Hardware zoom: the loop's own transient is shorter than one sample
VM2-VM3 [V]
Time from event [s]